

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2291

Tristate 345 kV
December 24, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2291. The Interconnection Customer proposes a 155.3 MW solar plus storage facility interconnecting at the Tristate 345 kV substation in the West Virginia area. The request was received on February 21, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 480 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$904,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Wiccopee / N Madison 345 138kV
Contingency: Valemount / Pence 345kV Line
Loading: 106.5% (Rating: 240 MVA)
DFAX: 5.6%
Pre-project loading: 102.9%
- Facility: Tackett Crk / Wiccopee 138kV
Contingency: Pence / Valemount Substation 345kV Line
Loading: 105.8% (Rating: 234 MVA)
DFAX: 5.6%
Pre-project loading: 102.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wiccopee / N Madison 345 138kV	Reconductor line	\$769,000	25% (\$192,000)
Tackett Crk / Wiccopee 138kV L	Reconductor line	\$2,849,000	25% (\$712,000)
	Total allocated to project		\$904,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.